

SAPHUL PMS

Database Tables:

PK: Primary Key

FK : Foreign Key

1) USERS

Purpose: Store all system users (Admin, Project Managers, Members)

Column	Type	Description
id	INT (PK)	Unique user ID
name	VARCHAR(100)	Full name
email	VARCHAR(150)	Unique login
role	ENUM	ADMIN / PROJECT_MANAGER / MEMBER
status	ENUM	ACTIVE / INACTIVE
is_deleted	BOOLEAN	Soft delete
deleted_at	TIMESTAMP	Deletion time
created_at	TIMESTAMP	Created
updated_at	TIMESTAMP	Updated

2) PROJECTS

Purpose: Store projects.

Completion requires Admin approval.

Column	Type	Description
id	INT (PK)	Project ID
name	VARCHAR(150)	Project name
description	TEXT	Project details
status	ENUM	PENDING / IN_PROGRESS / COMPLETED / DELAYED
review_status	ENUM	NONE / UNDER_REVIEW / APPROVED / REJECTED
start_date	DATE	Start date
end_date	DATE	Planned end
review_requested_at	TIMESTAMP	Sent for admin review
reviewed_at	TIMESTAMP	Approved time
reviewed_by	FK → users.id	Admin
created_by	FK → users.id	Admin
is_deleted	BOOLEAN	Soft delete
deleted_at	TIMESTAMP	Deletion time
created_at	TIMESTAMP	Created
updated_at	TIMESTAMP	Updated

3) PROJECT_MEMBERS

Purpose:

Defines which users belong to which projects and in what role.

Supports multiple Project Managers and members across multiple projects.

Column	Type	Description
id	INT (PK)	Row ID
project_id	FK → projects.id	Project
user_id	FK → users.id	User
role	ENUM	PROJECT_MANAGER / MEMBER
assigned_at	TIMESTAMP	Assignment time
is_deleted	BOOLEAN	Soft delete
deleted_at	TIMESTAMP	Removal time

4) TASKS

Purpose:
Store tasks.
Completion requires Project Manager approval.

Column	Type	Description
id	INT (PK)	Task ID
project_id	FK → projects.id	Parent project
title	VARCHAR(150)	Task title
description	TEXT	Task details
assigned_to	FK → users.id	Assigned member
created_by	FK → users.id	Admin / PM
status	ENUM	PENDING / IN_PROGRESS / COMPLETED
review_status	ENUM	NONE / UNDER_REVIEW / APPROVED / REJECTED
priority	ENUM	LOW / MEDIUM / HIGH
start_date	DATE	Start
due_date	DATE	Deadline
review_requested_at	TIMESTAMP	Sent for PM review
reviewed_at	TIMESTAMP	Approved
reviewed_by	FK → users.id	Project Manager
completed_at	TIMESTAMP	Final completion
is_deleted	BOOLEAN	Soft delete
deleted_at	TIMESTAMP	Deletion time
created_at	TIMESTAMP	Created
updated_at	TIMESTAMP	Updated

5) PROJECT_FILES

Purpose:

Store file metadata only.

Actual files live in cloud storage (AWS S3 / GCP / Azure/local storage).

Column	Type	Description
id	INT (PK)	File ID
project_id	FK → projects.id	Project
task_id	FK → tasks.id (nullable)	Task
uploaded_by	FK → users.id	Uploader
file_name	VARCHAR(255)	File name
file_type	VARCHAR(50)	pdf / docx / png
file_url	TEXT	Storage path
version	VARCHAR(20)	v1, v2, v3
is_latest	BOOLEAN	Latest version
is_deleted	BOOLEAN	Soft delete
deleted_at	TIMESTAMP	Deletion time
created_at	TIMESTAMP	Uploaded

6) ACTIVITY_LOGS

Purpose:

Audit trail & timeline for all important actions.

Column	Type	Description
id	INT (PK)	Log entry ID
entity_type	ENUM	PROJECT / TASK / FILE
entity_id	INT	ID of entity
action	VARCHAR(50)	created / updated / sent_for_review / approved
performed_by	FK → users.id	Actor
created_at	TIMESTAMP	Time

7) PROJECT_TASK_STATS

Purpose:
Performance & dashboard optimization (summary table).

Column	Type	Description
project_id	PK / FK → projects.id	Project
total_tasks	INT	Total tasks
completed_tasks	INT	Completed
under_review_tasks	INT	Awaiting review
delayed_tasks	INT	Overdue
last_updated	TIMESTAMP	Sync time

RELATIONSHIPS (among tables) :

Relationship	Type
Users ↔ Projects	Many-to-Many (project_members)
Projects → Tasks	One-to-Many
Users → Tasks	One-to-Many
Projects → Files	One-to-Many
Tasks → Files	One-to-Many
Users → Activity Logs	One-to-Many

INDEXES (for fast search in the tables):

Table	Indexed Columns
project_members	project_id, user_id
tasks	project_id
tasks	assigned_to
tasks	status
tasks	review_status
projects	status
projects	review_status
project_files	project_id